



PROJECT TITLE -

**SEASONAL AGRICULTURE
PERFORMANCE ANALYSIS**

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PROBLEM STATEMENT

Agricultural activities face **huge performance variations** due to seasonal changes in environment and markets.

- **Seasonal Variations:** Conditions fluctuate dramatically across seasons.
- **Performance Fluctuations:** Farm success differs between cycles.
- **Data Limitations:** Raw numbers hide critical trends.
- **Analytical Goal:** Uncover patterns using data analytics.



PROJECT DESCRIPTION

- **Core Dataset:** Farm-level records of 4,000 observations across 28 distinct features.
- **Features Analysed:**
 - **Environmental:** Rainfall, temperature, soil moisture, and pH.
 - **Operational:** Fertilizer/pesticide inputs, water usage, and seed quality.
 - **Financial:** Crop yields, market price, total cost, revenue, and net profit.
- **Analytical Focus:** Exploratory data analysis (EDA) and bivariate profiling to uncover hidden seasonal trends without using machine learning.

WHO ARE THE END USERS?



- **Farmers & Cooperatives:** Helping them make informed decisions on resource allocation based on seasonal trends.
 - **Agronomists & Researchers:** Aiding their understanding of crop vulnerabilities to seasonal weather shifts.
 - **Agricultural Policymakers:** Providing data-backed insights for planning regional support, subsidies, and market interventions.
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TECHNOLOGY USED

- **Google Colab:** Cloud-based Jupyter Notebook environment used for executing the entire data analytics workflow.
- **Python:** Primary programming language used for data manipulation, statistical analysis, and plotting.
- **Pandas & NumPy:** Core libraries applied for dataset cleaning, missing value imputation, and aggregating 4,000 farm records.
- **Seaborn & Matplotlib:** Visualization tools used to design clean statistical charts like seasonal box plots, heatmaps, and counts.



RESULTS

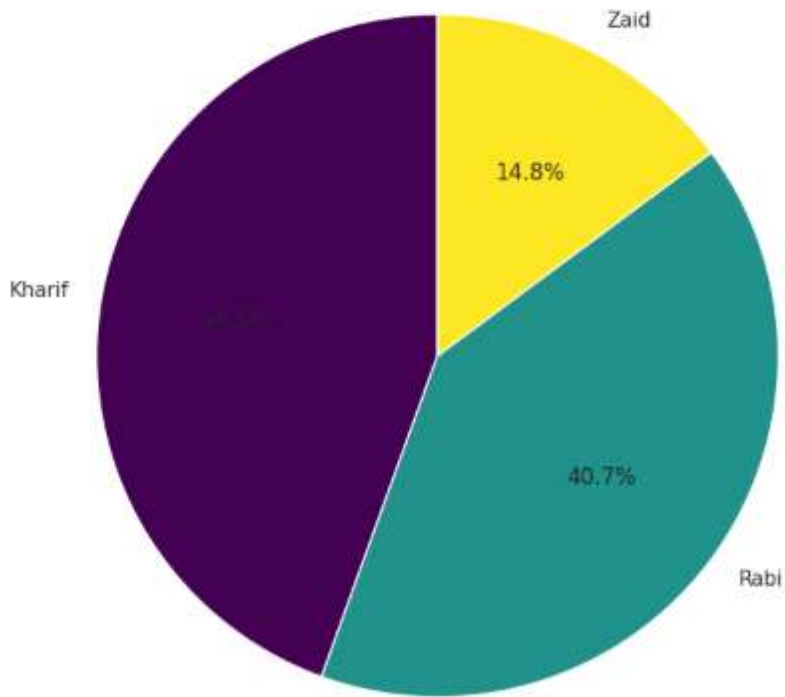
- **Kharif Season Highs:** Records the highest financial returns with an average gross revenue of ₹710,719 and a net profit of ₹178,914.
- **Rabi Season Stability:** Represents a stable, reliable farming cycle characterized by a moderate average net profit of ₹87,689.
- **Zaid Season Deficit:** Experiences severe operational and financial strain, resulting in an average net loss of ₹24,804 per farm.
- **Profitability Drivers:** Bivariate analysis proves net profit is limited more by high seed costs and market price shifts than fertilizer inputs.

RESULTS

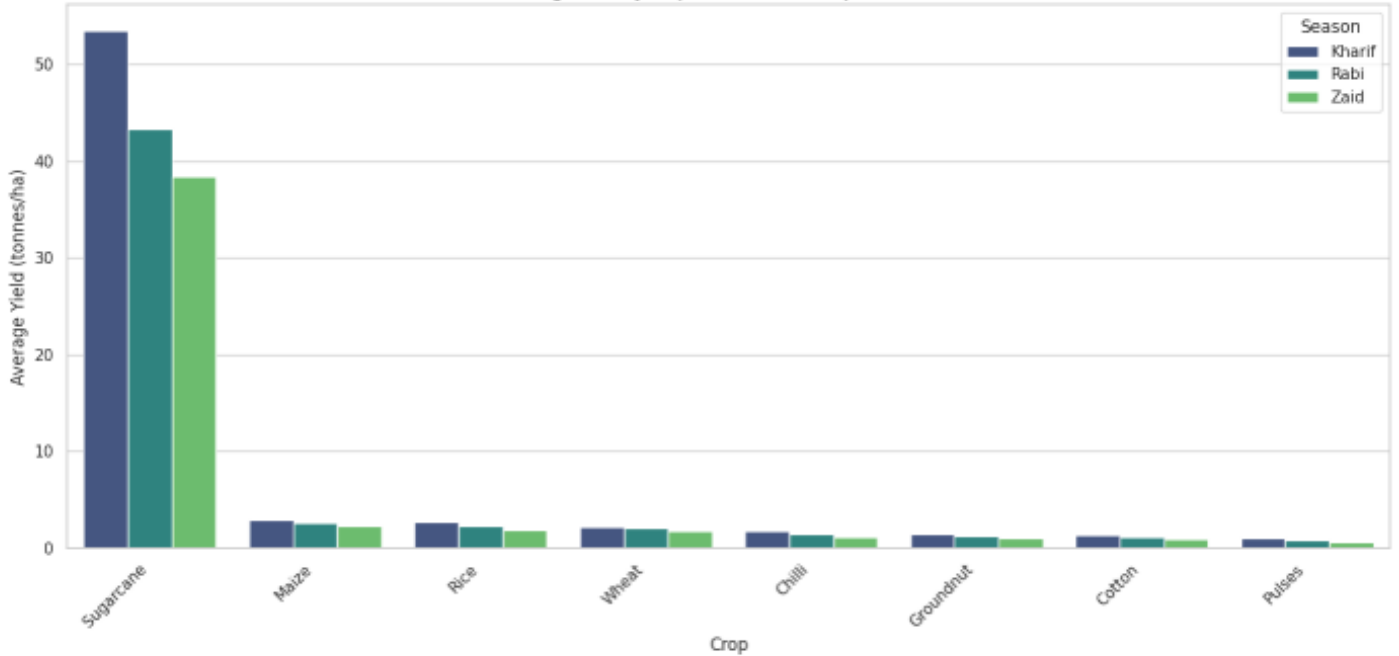
- **Zaid Resource Stress:** Suffers from severe water inefficiency, requiring a massive average irrigation volume of **6,419.89 m³**.
- **Kharif Disease Exposure:** Faces the highest environmental vulnerability with pest and disease risk thresholds spiking to **54.47%**.
- **Soil Health Balance:** Excess chemical fertilizer use fails to boost crop yields and actively degrades baseline soil pH levels.
- **Environmental Limits:** High summer temperatures during Zaid (averaging **31.04 °C**) directly drive down soil moisture to a low **19.28%**.

RESULTS

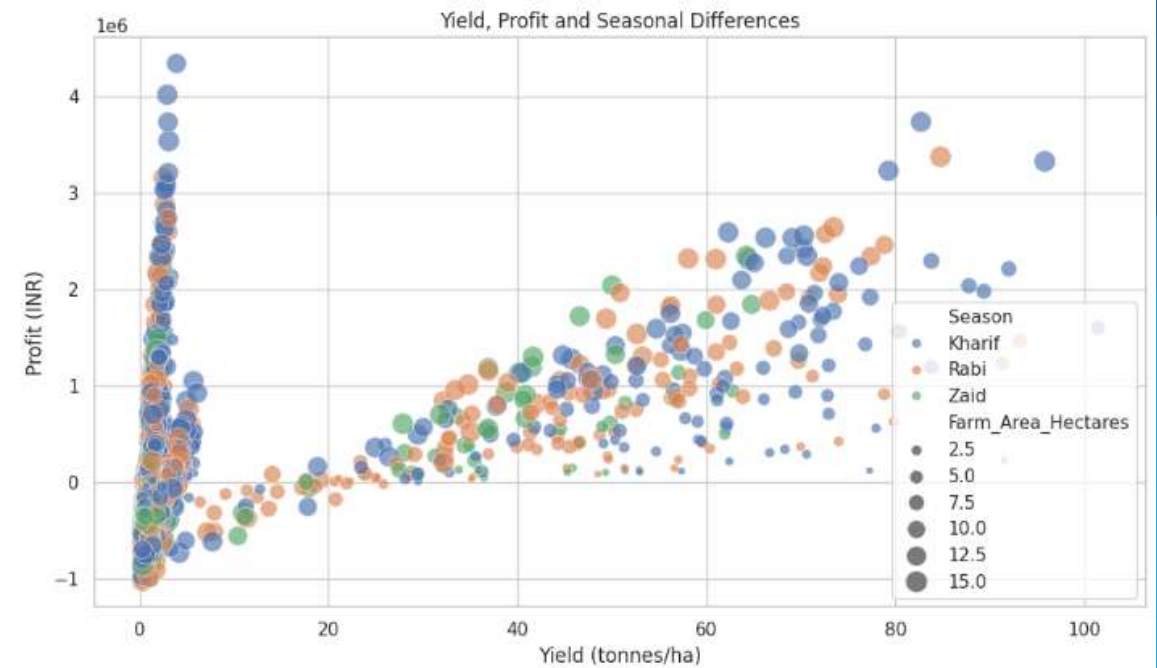
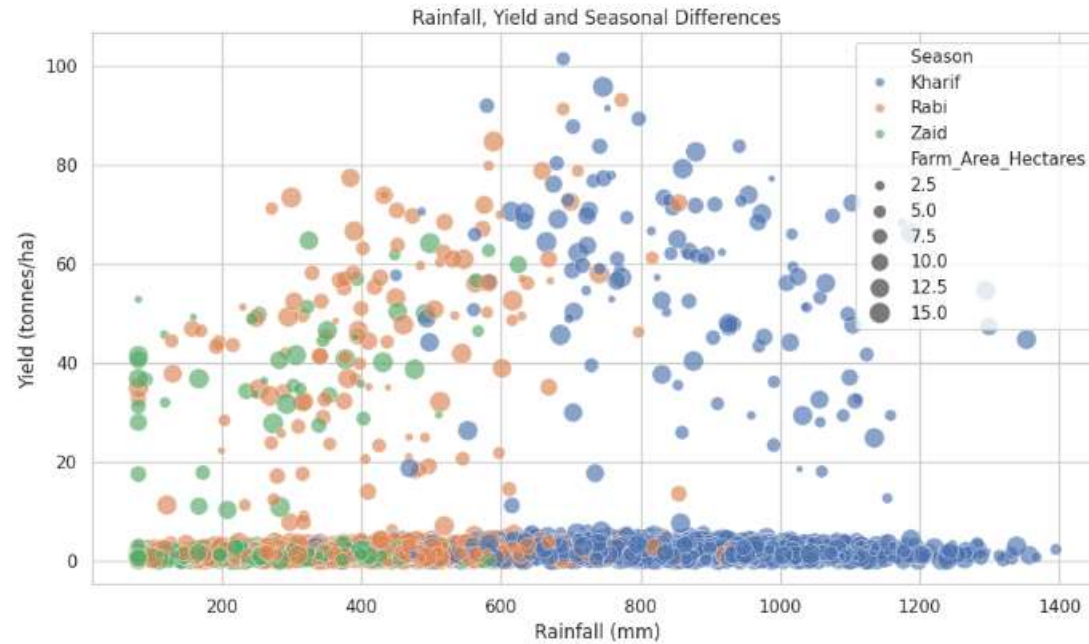
Distribution of Records by Season



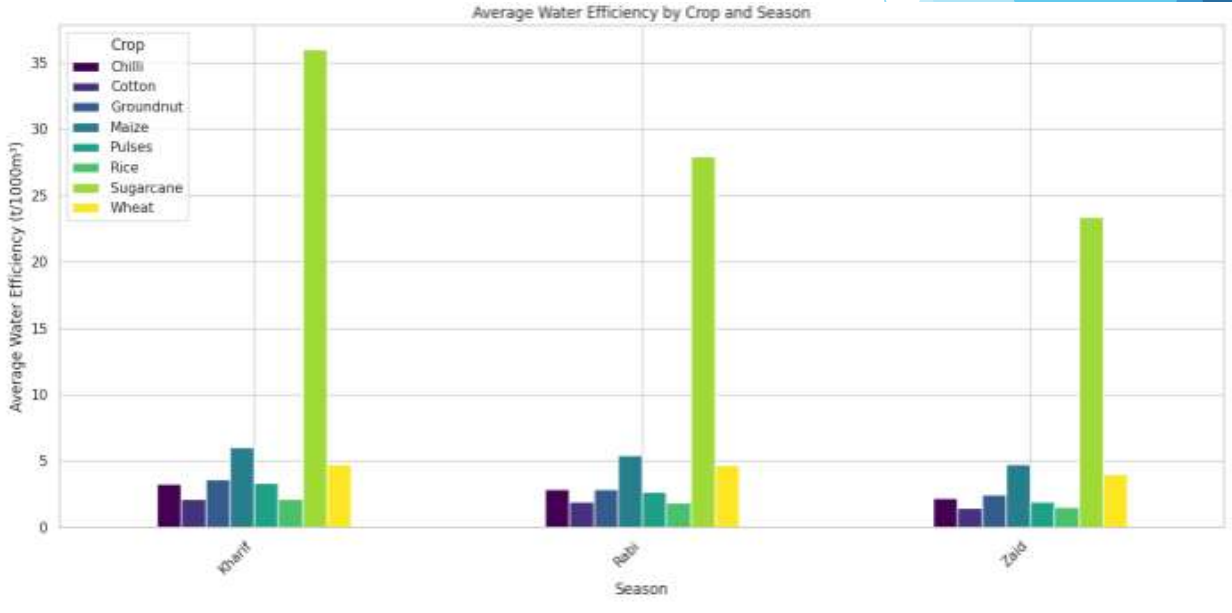
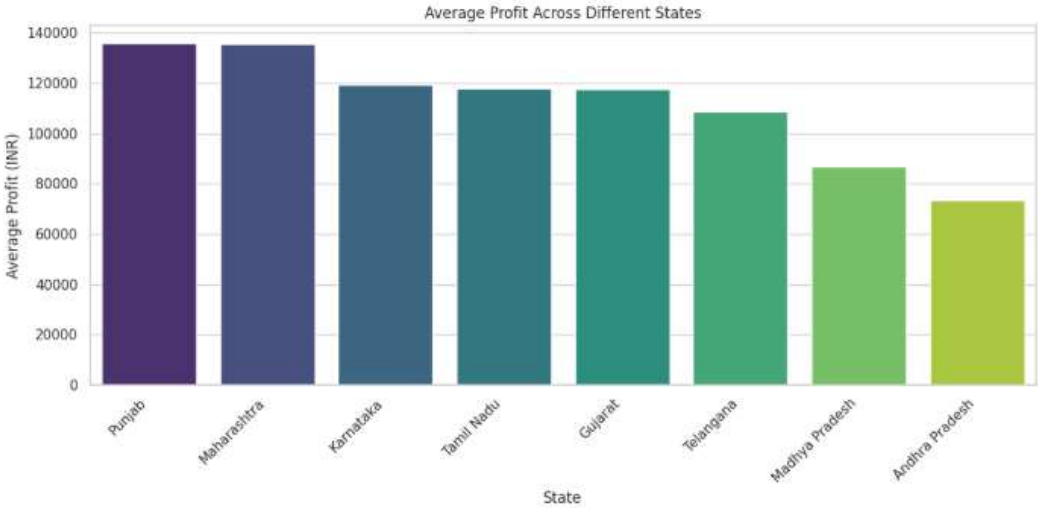
Average Yield by Crop and Season (Grouped Bar Plot)



RESULTS



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FUTURE SCOPE

- **Predictive Yield Modeling:** Apply machine learning regressors to forecast future crop yields based on environmental factors.
- **Interactive Analytics Dashboard:** Develop a web-based Streamlit application to display real-time seasonal insights and farm trends.
- **Resource Optimization Engine:** Design an automated system to provide targeted fertilizer and irrigation recommendations using soil data.
- **Early Disease Warning System:** Create anomaly detection triggers to flag high pest and disease risk periods automatically.

GITHUB LINK

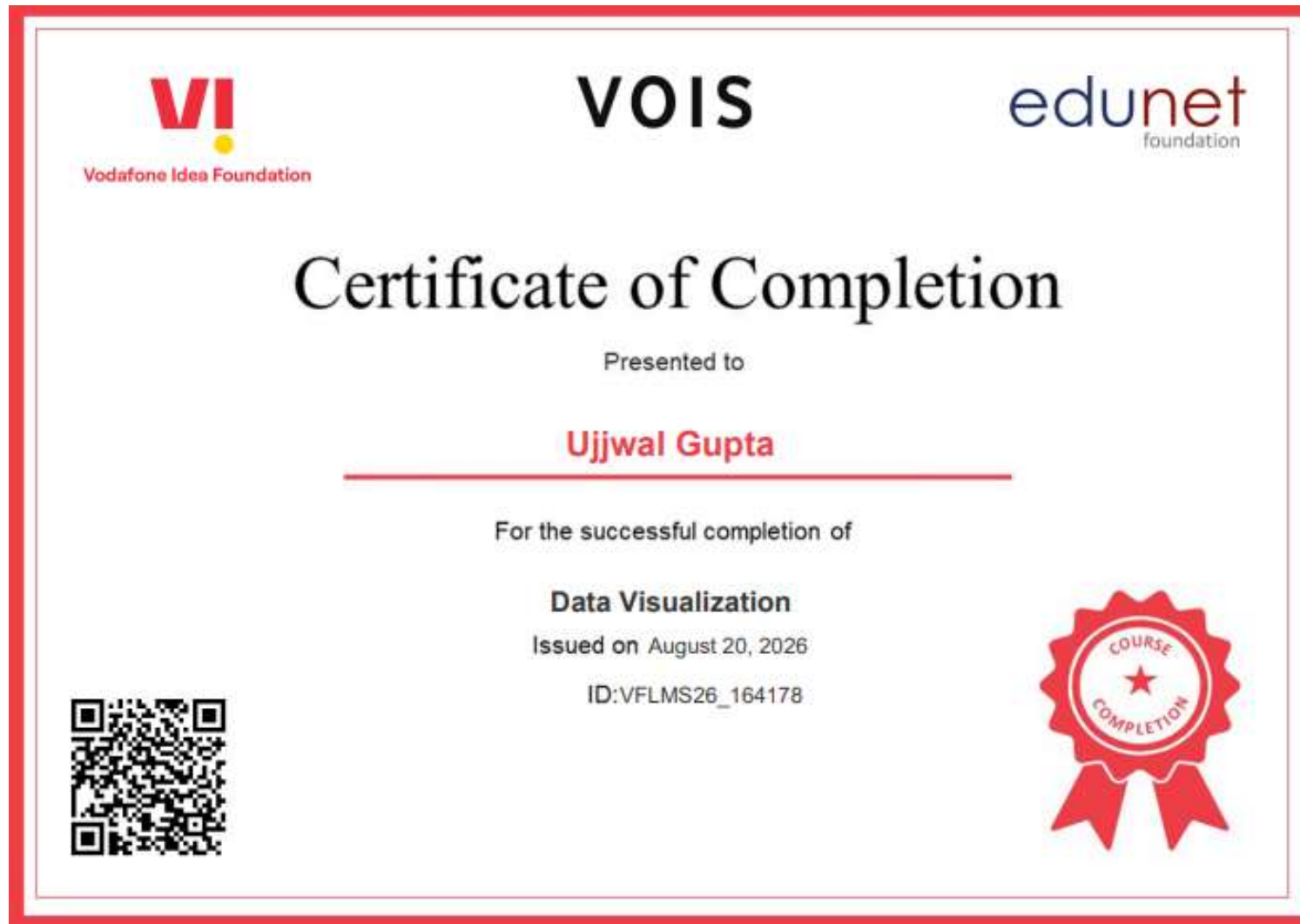


<https://github.com/ujjwalgupta2021/VOIS-Data-Analytics/tree/main/Major%20Project-Seasonal%20Agriculture%20Performance%20Analysis>

[Click Here](#)



VOIS COURSE COMPLETION CERTIFICATE - COURSE NAME- DATA VISUALIZATION





THANK YOU

